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FORM TP 2014178

MAY/JUNE 2014

CARIBBEAN EXAMINATIONS COUNCIL
CARIBBEAN ADVANCED PROFICIENCY EXAMINATION®
ENVIRONMENTAL SCIENCE
ECOLOGY, PEOPLE AND NATURAL RESOURCE USE

UNIT 1 – Paper 02

2 hours 30 minutes

06 MAY 2014 (p.m.)

READ THE FOLLOWING INSTRUCTIONS CAREFULLY.

1. This paper consists of SIX questions, TWO from each module.
2. Candidates MUST answer ALL questions.
3. Write your answers in the answer booklet provided.
4. The use of a silent, non-programmable, scientific calculator is permitted.

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02120020/CAPE 2014

MODULE 1

Answer BOTH questions.

1. (a) Figure 1 shows the different trophic levels in a terrestrial ecosystem. The total amount of usable energy present at the level of the producers is 30 000 kilocalories.

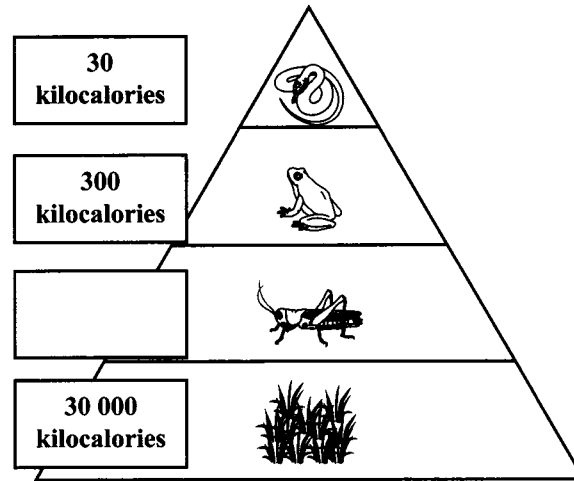


Figure 1. Trophic levels in a terrestrial ecosystem

- (i) Calculate the amount of energy that will be present at the primary consumer level. [2 marks]
- (ii) Explain the pattern of energy flow through the trophic levels. [2 marks]
- (b) (i) Define the term 'carrying capacity'. [2 marks]
- (ii) Why is carrying capacity important in maintaining balance in ecosystems? [4 marks]
- (c) Explain why phosphorous is a limiting growth factor in terrestrial ecosystems. [4 marks]

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- (d) A biologist attempted to estimate the population of deer in Royal Park using the capture-mark-recapture technique as shown below.

TABLE 1: STEPS IN THE CAPTURE-MARK-RECAPTURE TECHNIQUE USED TO ESTIMATE THE POPULATION OF DEER IN ROYAL PARK

Step 1	A biologist caught and marked 100 deer from Royal Park.
Step 2	The 100 marked deer were released back into their habitat at Royal Park.
Step 3	The biologist recaptured 55 deer. Out of this number, 25 were from the original marked sample.

- (i) Based on the information provided in Table 1, use an appropriate formula to calculate the estimated population size of deer in Royal Park. **[4 marks]**
- (ii) State TWO assumptions of the method used by the biologist. **[2 marks]**

Total 20 marks

2. (a) Figure 2 is a graph which represents a predator–prey relationship. Study Figure 2 and answer the questions that follow.

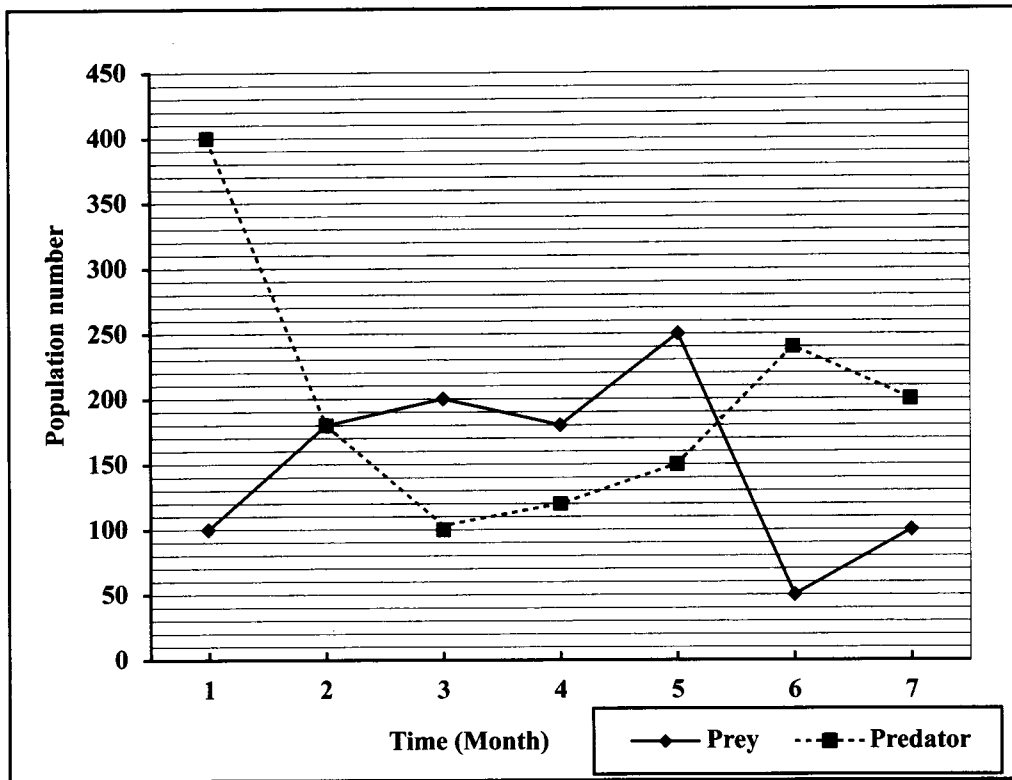


Figure 2. Graph representing a predator–prey relationship

- (i) Explain the trend in the population numbers of the predator and prey as shown in the graph. **[3 marks]**
- (ii) Calculate the change in the size of the predator population between Month 2 and Month 7. **[3 marks]**
- (b) Explain the importance of natural selection in the evolution of species. **[4 marks]**
- (c) Explain TWO ways by which human activities can disrupt the integrity of natural ecosystems. **[4 marks]**

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- (d) Figure 3 shows data collected from an ecological study of a school garden.

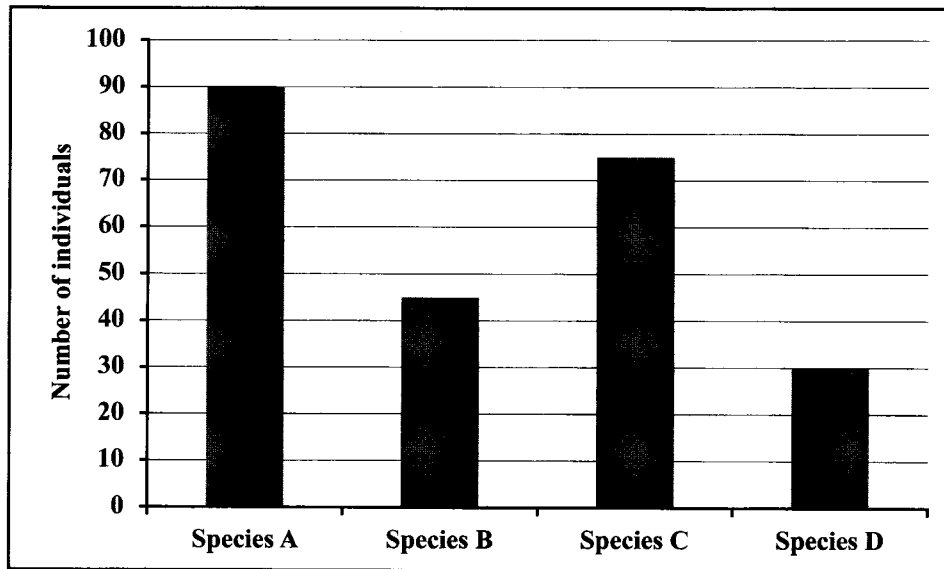


Figure 3. Data collected from an ecological study

- (i) Use the data in Figure 3 to calculate the TOTAL number of individuals of the TWO LEAST collected species. **[2 marks]**
- (ii) Using an appropriate formula, calculate the species diversity for the school garden. **[4 marks]**

Total 20 marks

MODULE 2

Answer BOTH questions.

3. (a) Table 2 shows data on the factors that are used to calculate the Human Development Index (HDI) for two countries, A and B, for the year 2011.

TABLE 2: FACTORS USED TO CALCULATE HUMAN DEVELOPMENT INDEX FOR COUNTRIES A AND B

Country	Health	Education	Life Expectancy
A	0.664	0.406	62.1
B	0.964	0.985	81.1

- (i) What is meant by the term 'Human Development Index' (HDI)? [1 mark]
- (ii) Make FOUR deductions from the data presented in Table 2. [4 marks]
- (b) "Educating women is a critical approach that can be used by countries for achieving the goal of sustainable development." Evaluate this statement giving FOUR points in your response. [4 marks]
- (c) Figure 4 shows the per capita freshwater consumption of a country from 1999 to 2004.

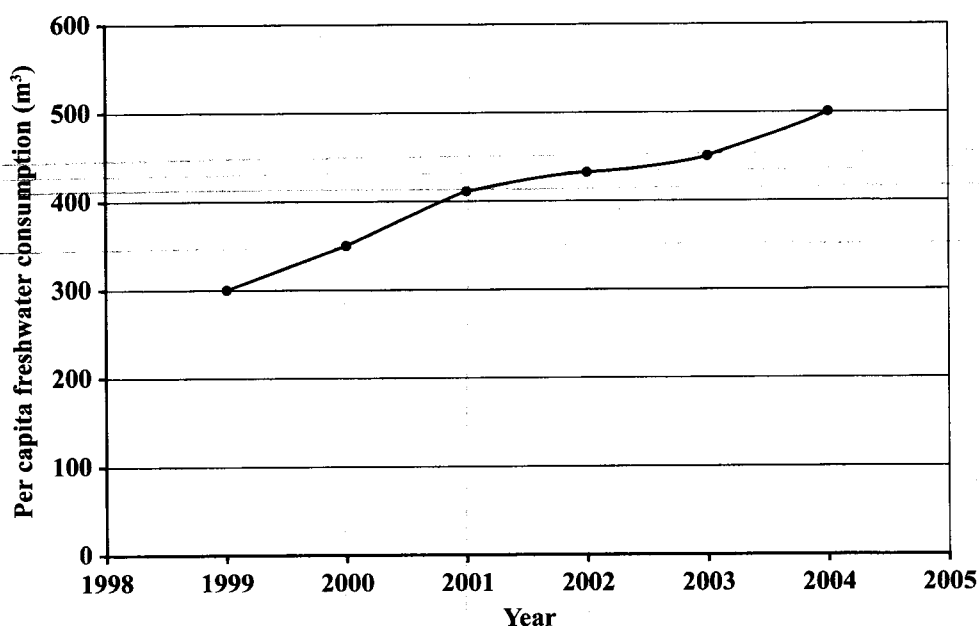


Figure 4. Per capita freshwater consumption (m³) of a country from 1999 to 2004

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- (i) State the meaning of the term 'per capita freshwater consumption'. [2 marks]
- (ii) Calculate the rate of per capita consumption of freshwater for the period 2000 to 2003. [3 marks]
- (iii) Suggest a plausible reason for the trend in per capita freshwater consumption shown in Figure 4. [2 marks]
- (iv) Describe TWO possible environmental impacts on the freshwater resource which could result from the trend in consumption shown in Figure 4. [4 marks]

Total 20 marks

4. (a) Table 3 provides data on infant mortality for two countries, A and B.

**TABLE 3: INFANT MORTALITY RATE (PER THOUSAND)
FOR COUNTRIES A AND B**

Country	Infant Mortality Rate
A	16.4
B	29.7

- (i) Define the term 'infant mortality'. [1 mark]
- (ii) Giving a reason for your answer, state which country will have a FASTER growing population. [2 marks]
- (iii) Giving a reason for your answer, state which country is most likely a developing country. [2 marks]
- (b) Table 4 presents some demographic data for a country for three different years.

TABLE 4: DEMOGRAPHIC DATA FOR A COUNTRY

	Birth Rate	Death Rate	Annual Population Growth Rate
1990	4.8%	2.2%	
2000	2%	0.9%	1.1
2010	1.5%	0.7%	0.8%

- (i) What is meant by the term 'annual population growth rate'? [1 mark]
- (ii) Use the data provided in Table 4 to calculate the annual population growth rate for the country in the year 1990. [2 marks]
- (iii) Calculate the doubling time for the country for the year 2010. [3 marks]
- (iv) State the trend in annual population growth rate and describe ONE implication that it may have for the country. [3 marks]
- (c) Discuss TWO impacts that increasing population growth would have on less developed countries. [6 marks]

Total 20 marks

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MODULE 3

Answer BOTH questions.

5. (a) State TWO ecological reasons and ONE ethical reason for resource conservation. [3 marks]
- (b) Why is slash and burn agriculture as practised by Indigenous Caribbean Peoples considered a sustainable practice? [4 marks]
- (c) Table 5 shows the annual amount of municipal solid waste received by a landfill in a Caribbean country. Examine the data in Table 5 and answer the questions that follow.

**TABLE 5: ANNUAL MASS OF MUNICIPAL SOLID WASTE RECEIVED
BY A LANDFILL IN A CARIBBEAN COUNTRY**

Year	1996	1997	1998	1999	2000
Mass of Solid Waste (tonne)	133 700	177 400	176 900	172 300	196 000
Year	2001	2002	2003	2004	2005
Mass of Solid Waste (tonne)	207 500	238 500	269 400	298 500	

- (i) What is the TOTAL mass of solid waste sent to the landfill from 1998 to 2000? [2 marks]
- (ii) Estimate the annual mass of solid waste that will be sent to the landfill in the year 2005. [2 marks]
- (iii) State the trend highlighted by Table 5, and explain how a recycling scheme might help the problem illustrated in Table 5. [3 marks]
- (d) (i) Explain how natural resource conservation can be achieved by EITHER land use planning and zoning regulations OR sustainable yield management. [3 marks]
- (ii) Suggest THREE systems that the government could use to effectively achieve natural resource conservation in Caribbean countries. [3 marks]

Total 20 marks

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6. (a) Why can fertile soil, a natural resource, be described as BOTH renewable AND exhaustible? [2 marks]
- (b) State ONE consumptive and TWO non-consumptive uses of natural resources. [3 marks]

Figure 5 illustrates the average net productivity of some natural ecosystems. Study Figure 5 and answer the questions that follow.

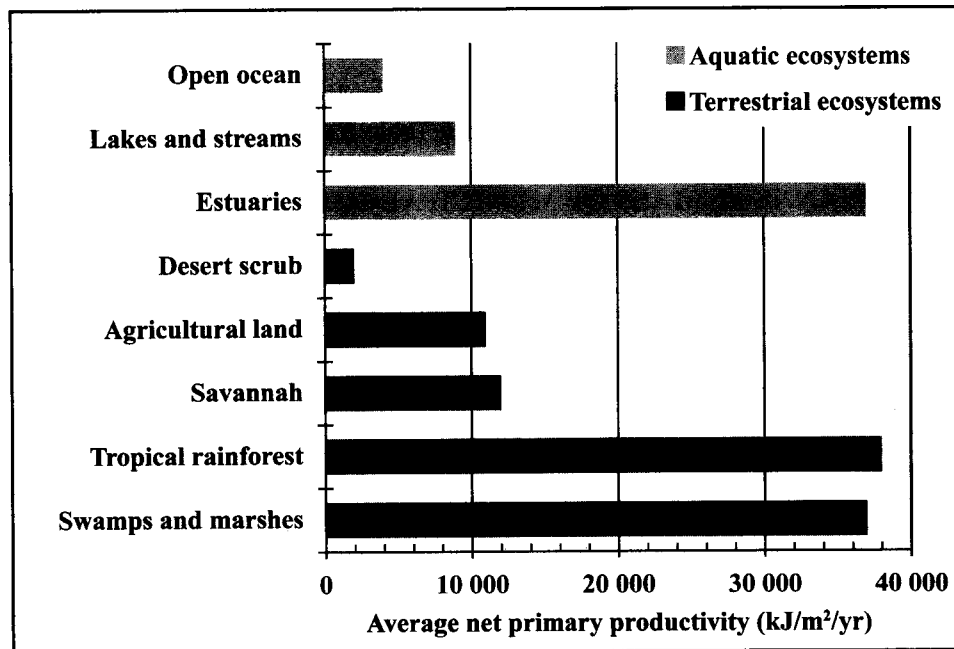


Figure 5. Average net primary productivity in some natural ecosystems

- (c) (i) What is the total net primary productivity that could be achieved from the tropical rainforest and savannah ecosystems? [3 marks]
- (ii) Which ecosystems have the HIGHEST and LOWEST average net primary productivity? [2 marks]
- (d) (i) Identify THREE **named** natural resources present in a **named** Caribbean country. [4 marks]
- (ii) Discuss TWO reasons for the importance of EACH of the natural resources identified in Part (d) (i) above. Each reason may be used only once. [6 marks]

Total 20 marks

END OF TEST

IF YOU FINISH BEFORE TIME IS CALLED, CHECK YOUR WORK ON THIS TEST.